

Noise Removal from a Real ECG Signal in Multisim

Design 1: Active High-Pass Stage + Passive Twin-T Notch

1 Objective

The aim was to take a real ECG recording, corrupt it with realistic noise inside Multisim, and recover the heartbeat with a filter chain. Two interference types were chosen: slow *baseline wander* (as caused by breathing and body movement) and *50 Hz mains hum*. This document covers **Design 1**, in which the hum is removed by a *passive* twin-T notch filter.

2 Signal Source

The signal is a real recording from the MIT-BIH Arrhythmia Database (record 100, lead MLII, sampled at 360 Hz), obtained as a CSV file. The CSV stores raw ADC counts, which were converted to millivolts with the database's standard rule, $(\text{count} - 1024)/200$. A clean 10 s segment (13 regular beats, starting at the 318 s mark) was cut out, shifted to a zero baseline and scaled so that the R-peaks reach about 1 V. This gives a heart rate near 80 bpm. The result was written as a two-column **time voltage** text file and loaded into a Multisim piecewise-linear (PWL) voltage source named *HeartBeat*. No signal shape was synthesized; only the units and amplitude were changed.

3 Noise Added

Noise was added by placing two further AC sources in series with the ECG source, so that the voltages add.

Source	Models	Amplitude	Frequency
BaselineWander	breathing / movement drift	$0.1 V_{\text{rms}}$	0.3 Hz
MainsHum	mains pick-up (India)	$0.05 V_{\text{rms}}$	50 Hz

The ECG content of interest lies roughly between 0.5 Hz and 40 Hz, so the wander sits just below the band and the hum just above it.

4 Schematic

Figure 1 shows the full circuit. The noisy signal passes through two stages in this order: a second-order Sallen-Key high-pass filter, then a twin-T notch filter.

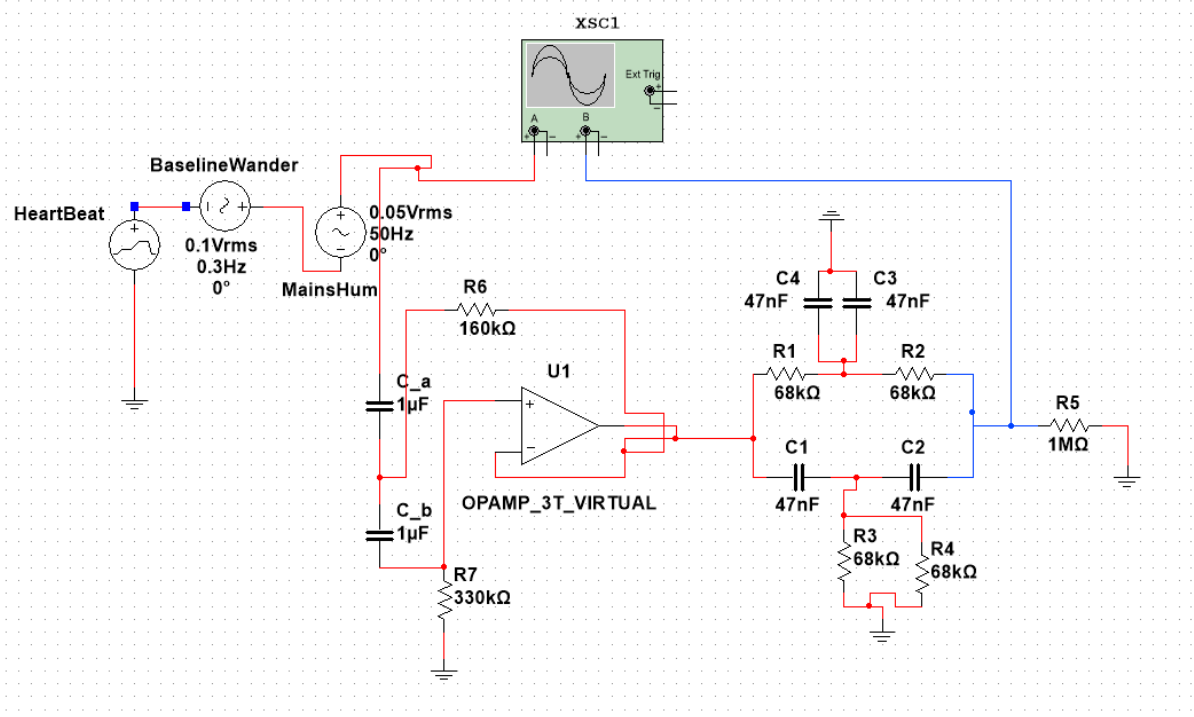


Figure 1: Design 1 schematic. Oscilloscope channel A is on the noisy signal, channel B on the filter output.

4.1 Stage 1: Sallen-Key high-pass (removes baseline wander)

The stage uses one op-amp as a unity-gain follower, two equal capacitors ($C_a = C_b = 1\mu\text{F}$), a feedback resistor $R_6 = 160\text{ k}\Omega$ from the junction of the capacitors to the op-amp output, and a resistor $R_7 = 330\text{ k}\Omega$ from the non-inverting input to ground. Its corner frequency and quality factor are

$$f_c = \frac{1}{2\pi C \sqrt{R_6 R_7}} \approx 0.69\text{ Hz}, \quad Q = \frac{1}{2} \sqrt{R_7 / R_6} \approx 0.72, \quad (1)$$

which gives a flat, Butterworth-like response above the corner and a 40 dB/decade roll-off below it.

4.2 Stage 2: passive twin-T notch (removes 50 Hz hum)

The twin-T is built from two parallel paths between its input and output. One path is R_1 – R_2 ($68\text{ k}\Omega$ each) with a capacitance of $2C$ ($C_3 \parallel C_4$, 94 nF) to ground from the middle. The other path is C_1 – C_2 (47 nF each) with a resistance of $R/2$ ($R_3 \parallel R_4$, $34\text{ k}\Omega$) to ground from the middle. At the notch frequency the signals through the two paths cancel:

$$f_0 = \frac{1}{2\pi RC} = \frac{1}{2\pi(68\text{ k}\Omega)(47\text{ nF})} \approx 49.8\text{ Hz}. \quad (2)$$

A $1\text{ M}\Omega$ resistor R_5 is connected from the output to ground as a light load.

4.3 Why this order

The Sallen-Key output is a low-impedance source, so it can drive the twin-T directly without loading it. Placing it first therefore removes the need for a separate buffer amplifier, and one op-amp is enough for the whole design. It also removes the large slow wander before the rest of the circuit sees it.

5 Results

5.1 Time domain (transient, 10 s)

Figure 2 compares the clean source ECG (channel A, red) with the filter output (channel B, blue). Figure 3 shows the noisy input (channel A, red) against the same output (channel B, blue).

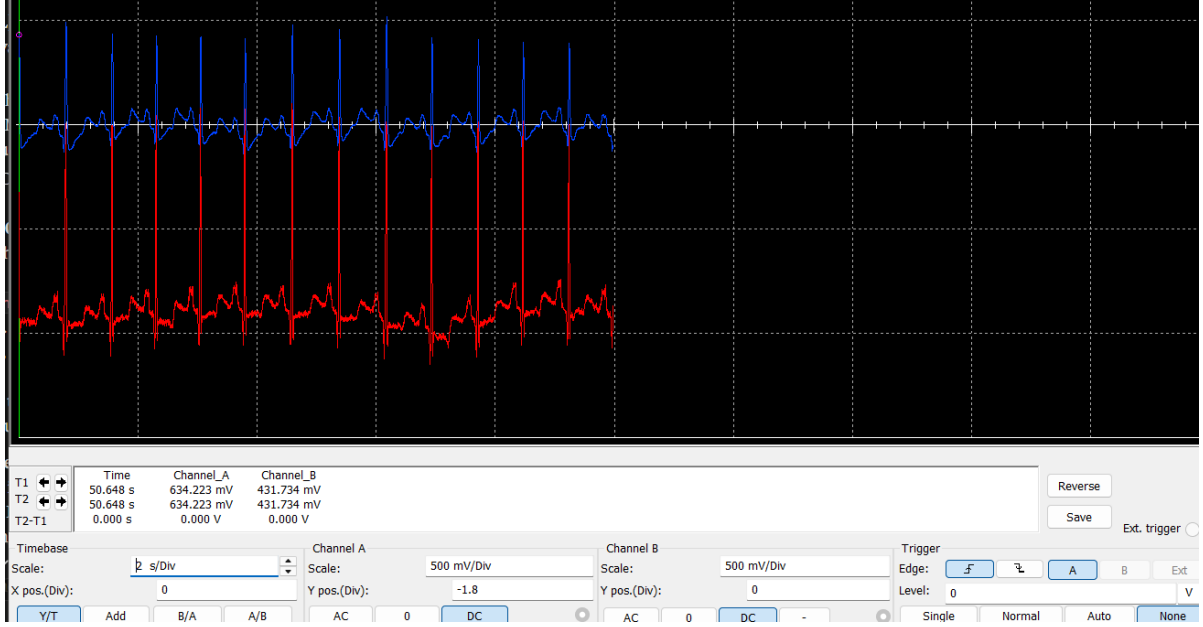


Figure 2: Clean ECG (red) and filter output (blue). The output keeps the R-peaks and the P and T waves.

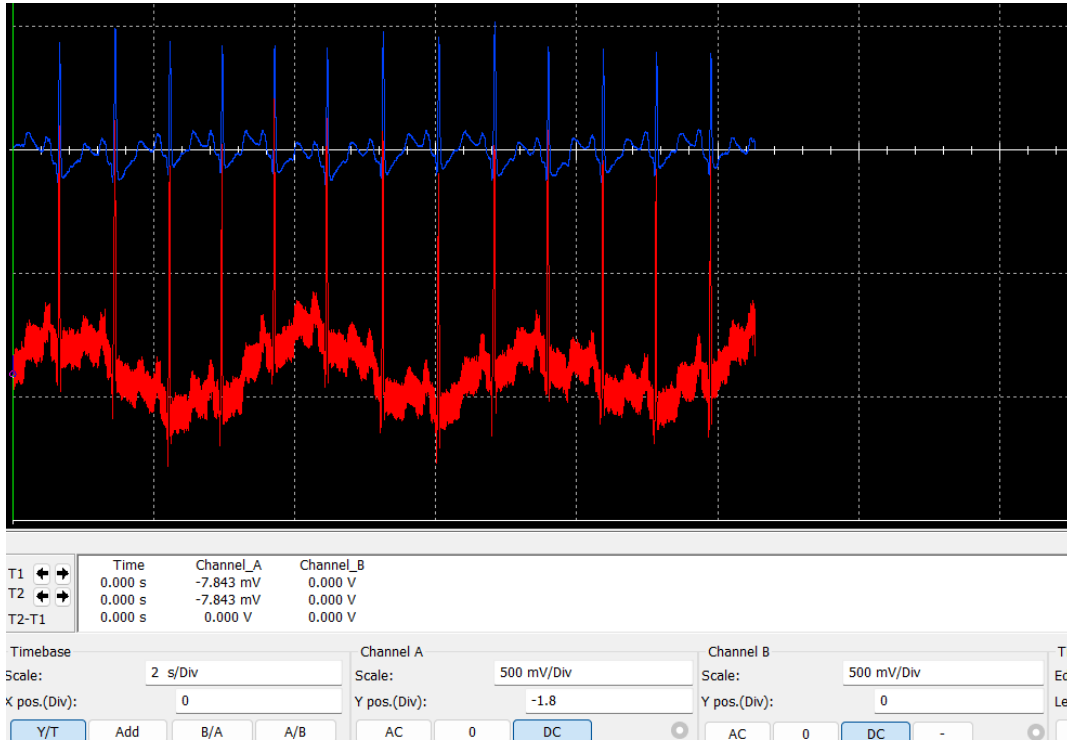


Figure 3: Noisy input (red) and filter output (blue). The input shows the slow wander and the thick 50 Hz band; the output shows neither.

In the noisy input the heartbeat is almost buried: the whole trace rises and falls slowly with the 0.3 Hz wander, and the 50 Hz hum forms a thick band around it. In the output the baseline is

flat, the hum fuzz is gone, and the R-peaks are clear, with the smaller P and T waves still visible between them. The wander is strongly reduced, though not completely removed (see the sweep results below).

5.2 Frequency domain (AC sweep)

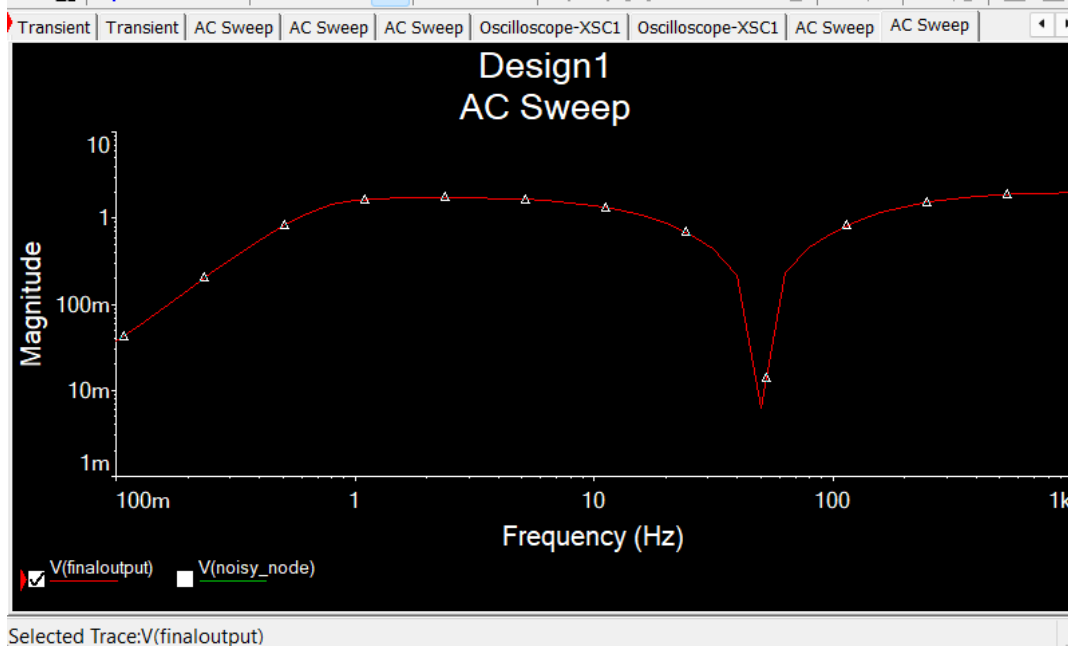


Figure 4: AC sweep of the filter output, 0.1 Hz to 1 kHz (log-log). The high-pass rise appears at low frequency and the notch near 50 Hz.

The sweep (Figure 4) shows the expected shape: a rising high-pass response below about 1 Hz, a flat passband, a narrow dip at 50 Hz, and recovery above it. Cursor readings were taken at four frequencies and are given below relative to the 1 Hz reading (1.602 V).

Frequency	Meaning	Output (V)	Relative gain	Comment
0.3 Hz	baseline wander	0.326	−13.8 dB	reduced to about 20%
1 Hz	reference	1.602	0 dB	—
40 Hz	top of ECG band	0.201	−18.0 dB	ECG content attenuated
49.5 Hz	mains hum	0.0075	−46.6 dB	about 0.5% remains

6 Discussion

What worked. Both noise sources were suppressed. The 50 Hz hum is attenuated by roughly 47 dB, so it has effectively disappeared from the output. The 0.3 Hz wander is reduced by about 14 dB, enough to flatten the baseline visibly. The heartbeat is clearly recoverable from a trace in which it was nearly hidden.

Limitation. A passive twin-T has a very wide, shallow notch (its quality factor is only about 0.25). Besides removing 50 Hz, it also attenuates the frequencies around it: the reading at 40 Hz is already −18 dB, and the response is falling well before that, as the sweep shows. The sharp QRS complex contains energy in exactly this region, so the R-peaks can come out slightly reduced and rounded. The R-peak height ratio between input and output was not measured here.

Possible improvement. Replacing the passive twin-T with an active one (feeding the R/2 and 2C arms from a buffered fraction of the output) narrows the notch considerably, which would protect the 20–45 Hz region while still removing the hum. This costs one additional op-amp and is the natural next design to compare against Design 1.

7 Summary

Signal	Real ECG, MIT-BIH record 100 (MLII), 10 s, 360 Hz, scaled to 1 V peak
Noise	0.3 Hz baseline wander + 50 Hz mains hum
Stage 1	Sallen-Key high-pass, $f_c \approx 0.69$ Hz, second order
Stage 2	Passive twin-T notch, $f_0 \approx 49.8$ Hz
Op-amps used	1
Hum suppression	about 47 dB
Wander suppression	about 14 dB at 0.3 Hz
Main drawback	wide notch also attenuates ECG content near 40 Hz
